AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 25 · STUDENT LESSON

# Generate, Retrieve, or Calculate?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

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## Before you start

Week 2: specify what output the task actually requires.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Distinguish retrieving a record, calculating from inputs and generating wording. Try the recall before opening notes.

### Your learning target

I can distinguish retrieving a record, calculating from **input**s and generating wording.

I can **trace** a task that combines these operations and check each step.

I can identify unsupported details and explain why **output** appearance does not reveal its method.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Different systems create, look up, or compute answers in different ways.



### Your target

LEARNING TARGET Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose **generation**, **retrieval** or **calculation** for each.

Which **output**s need checking?

Learner name:

### Readiness for the connected explanation

You need a chart of three fictional topic totals. Why should a named pupil spreadsheet not be the default **input**?

### This week's end goal

By the end, I can separate **retrieval**, **calculation** and **generate**d wording.

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STUDENT LESSON · 2 OF 18

# Notice the evidence

> Investigate the supplied Generate, Retrieve, or Calculate? case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

### Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



### Start with this puzzle

MYSTERY A story generator creates new wording, a library search **retrieve**s a stored title, and a calculator computes a result from stated operations.

**What do you notice?**

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

### Thinking habit

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

**A closer explanation**

Three tasks can produce similar-looking text while doing different work. **Retrieval** finds existing information. **Calculation** applies mathematical operations to **input** values.

**Generation** produces content using a model and the context it receives. A single application can combine all three, so the label AI does not tell you which step happened.

Our fictional makers club has a current notice N2: Wednesday, Room 4, six tables with four places at each table, and three places reserved for helpers. An older notice N1 says Room 2. N2 explicitly replaces N1 for this event.

These records are invented for this lesson; no search, account or real booking is needed.

To answer “Which room does the current notice list?”, find N2 and return Room 4 with the record identifier. That is **retrieval**. To answer “How many places remain after the helper reservation?”, use 6 × 4 − 3 = 21.

That is **calculation**. To suggest a friendly invitation, **generate** wording and check that its factual details agree with N2.

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STUDENT LESSON · 3 OF 18

## See how it works

> Follow the mechanism for Generate, Retrieve, or Calculate?. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

**Input → Process → Output**

Follow the information instead of imagining magic inside the machine.



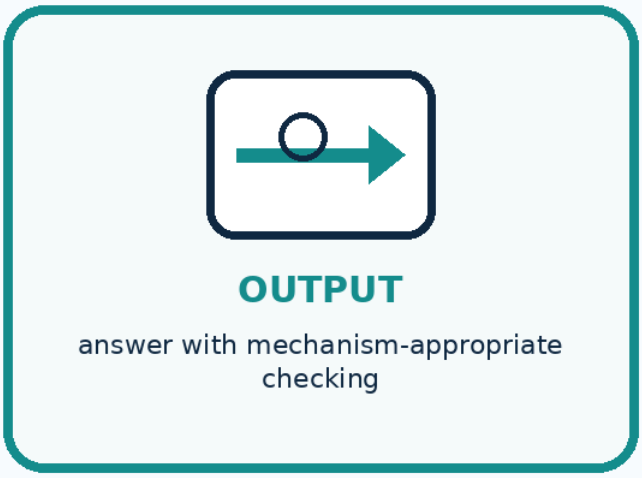
**INPUT**

question and available sources or rules



**PROCESS**

generate patterns, retrieve records, or calculate




**INPUT** question and available sources or rules

**PROCESS generate** patterns, **retrieve** records, or **calculate**

**OUTPUT** answer with **mechanism**-appropriate checking

### Key idea

CORE IDEA Different systems create, look up, or compute answers in different ways.

### How the mechanism works

UNDER THE HOOD **Generation** synthesizes **output**s from learned distributions, **retrieval** selects stored records, and **calculation** applies formal operations.

Products often combine all three; verification should match the component that produced each claim.

Draw a fresh example with all three stages:

## A closer explanation

### One club task with three kinds of work

| OPERATION | EXAMPLE OUTPUT | REQUIRED CHECK |
| --- | --- | --- |
| Retrieve N2 | Room 4 on Wednesday | Right event and version |
| Calculate | 6 × 4 − 3 = 21 | Inputs and operations |
| Generate wording | Try a making activity | Facts and constraints |

Retrieval supplies recorded information; calculation derives a result; generation drafts wording. Each operation needs a check suited to what it does.

Retrieving N1 accurately would still give an outdated room for this task. Successful **retrieval** means finding a record, not proving that every record is correct or current. Check the event, source and version.

Here the explicit replacement statement tells us which notice governs the classroom example.

A calculator given 6 × 4 correctly returns 24, but 24 is the total capacity, not the number remaining after three reserved places. Correct arithmetic on an incomplete expression answers the wrong question.

Explain the **input** values, units and operations before accepting a number.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## Compare the cases

> Compare the supplied cases for Generate, Retrieve, or Calculate?. Identify the detail that changes the conclusion.

PICTURE GALLERY

### Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.












### The case to investigate

CASE 1 — Invent a story Clue: new wording from patterns. Meaning: **generate**.

### The case to investigate

CASE 2 — Find a book title Clue: return catalog record. Meaning: **retrieve**.

### The case to investigate

CASE 3 — Add 27 + 18 Clue: apply arithmetic. Meaning: **calculate**.

### The case to investigate

CASE 4 — Answer with quote and total Clue: multiple components. Meaning: mixed system.

### Keep this boundary clear

BOUNDARY Fluent wording does not reveal whether a fact was **retrieve**d, **generate**d, or **calculate**d.

Which two pictures are easiest to confuse? What **evidence** separates them?

**Change one thing in the pictured case**

### Change one thing

Change: Reserve four places instead of two, keeping the room and day unchanged.

### Predicted result

**Prediction**: Eight learner places remain.

### Why the result follows

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

## Words that unlock the idea

> Use the vocabulary of Generate, Retrieve, or Calculate? to explain a concrete decision, rather than only memorising definitions.

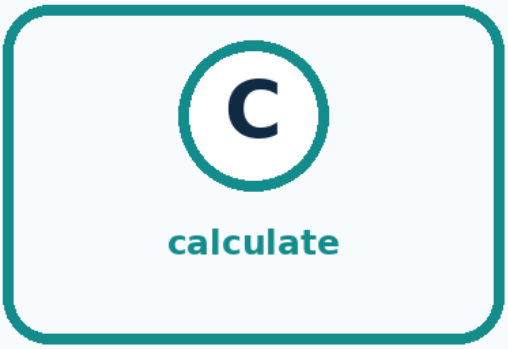
VOCABULARY

### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









| Word | Meaning | My example |
|---|---|---|
| generate | create a new output from learned patterns | |
| retrieve | find and return stored information | |
| calculate | apply defined operations to values | |
| provenance | where an output or claim came from | |

### Misconception detective

#### Find the unsupported claim

CLAIM TO REPAIR Any answer on a screen came from a database.

Use one vocabulary word and one piece of **evidence** to repair it:

> WORD POWER A strong definition helps you decide something in a new case.

### Words used in the closer explanation

| Word | Meaning |
| --- | --- |
| Retrieval | Finding and returning existing information from a collection |
| Calculation | Applying specified mathematical operations to input values |
| Generation | Producing content using a model and its input context |
| Source record | An existing item whose content and origin can be inspected |
| Trace | A record of the inputs, operations and outputs used in a task |

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STUDENT LESSON · 6 OF 18

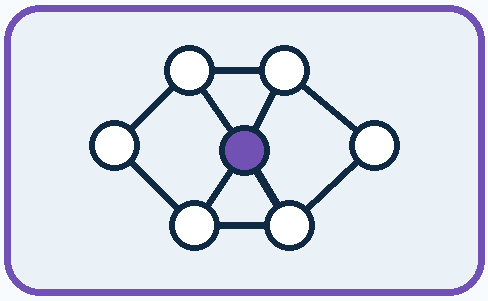
# Follow a worked example

> Follow the worked trace for Generate, Retrieve, or Calculate?, predict a step, then compare with its explanation.

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



### The case to investigate

CASE A story generator creates new wording, a library search **retrieve**s a stored title, and a calculator computes a result from stated operations.

### Step 1

STEP 1 — OBSERVE question and available sources or rules

### Step 2

STEP 2 — **TRACE generate** patterns, **retrieve** records, or **calculate**

### Step 3

STEP 3 — CONCLUDE answer with **mechanism**-appropriate checking

### Model answer

MODEL REASONING Check **generate**d claims against independent **evidence**, check **retrieve**d facts against the cited record and freshness, and recompute **calculation**s or inspect the operation. Mixed **output**s need more than one check.

### A claim to avoid

DO NOT OVERCLAIM Fluent wording does not reveal whether a fact was **retrieve**d, **generate**d, or **calculate**d.

What new **evidence** could change this answer?

### A closer explanation

| Draft statement | Evidence in this example | Decision |
|---|---|---|
| Meet Wednesday in Room 4 | Current notice N2 | Supported by the record |
| There are 21 non-helper places | 6 × 4 − 3 from N2 | Supported by calculation |
| Free snacks will be provided | Not stated in N2 | Remove or verify separately |

### Worked explanation · Generate, Retrieve, or Calculate?

A **generate**d invitation might add free snacks because that detail fits a familiar invitation pattern. Plausible wording is not **evidence** that snacks are provided.

Remove that claim or obtain an appropriate confirming source before including it. **Generate**d content can be useful and can contain correct facts; its fluent style alone does not verify those facts.

### Worked explanation · Generate, Retrieve, or Calculate?

**Generation** also does not guarantee originality. A response may repeat wording already present in its context or elsewhere.

> **Retrieval** and **calculation** can feed a generator, and a generator can propose a **calculation** for another tool. Keep the **evidence** path visible instead of assuming that every sentence came from the same operation.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Generate, Retrieve, or Calculate?

A fictional craft invitation combines three different operations.

### The facts we will use

> **Supplied facts**
>
> Approved record C: Tuesday, Hall 4, three tables with four learner places each.

> **Supplied facts**
>
> Two places are reserved for helpers; no snack information is supplied.

AI ACADEMY / AI-1/25

# Generate, Retrieve, or Calculate?

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Retrieve facts

Read Tuesday, Hall 4 and 3 x 4 places from approved C.



## 2. Calculate

12 total places minus 2 reserved =10 learner places.



## 3. Generate wording

Draft a friendly invitation using only those supported details.



## 4. Check claims

A sentence promising snacks has no supporting record.

**CHANGE ONE THING**

Reserve four places instead of two, keeping the room and day unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Retrieve facts | Read Tuesday, Hall 4 and 3×4 places from approved C. |
| 2. Calculate | 12 total places minus 2 reserved =10 learner places. |
| 3. Generate wording | Draft a friendly invitation using only those supported details. |
| 4. Check claims | A sentence promising snacks has no supporting record. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

**Retrieval** copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

### What this example does not establish

#### Keep the stated limit

The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

#### Keep the stated limit

Read the labelled picture: Label each arrow with its operation. Can the system add snacks because invitations often mention them?

#### Keep the stated limit

Change one thing: Reserve four places instead of two, keeping the room and day unchanged. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Generate, Retrieve, or Calculate? claim and explain a repair using the supplied evidence.

COMPARE

### Same Result, Different Reason

A comparison helps reveal the true concept boundary.

| Case | Evidence | What it shows |
| --- | --- | --- |
| Invent a story | new wording from patterns | generate |
| Find a book title | return catalog record | retrieve |
| Add 27 + 18 | apply arithmetic | calculate |
| Answer with quote and total | multiple components | mixed system |

### Repair two tempting claims

#### Challenge claim A

CLAIM A Any answer on a screen came from a database.

Repair:

#### Challenge claim B

CLAIM B **Calculate**d **output**s never need checking.

Repair:

#### A frame for explaining

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

### A closer explanation

#### Check the claim against the facts

For an existing meeting time, **retrieve** the relevant notice. For a total, **calculate** using the right **input**s and units. For a draft title or invitation, **generation** can offer wording, followed by review.

You can also do these classroom tasks on paper. Choose the operation because it fits the question, not because a tool is described as intelligent.

#### Check the claim against the facts

A complete answer keeps source facts, derived results and suggested wording distinguishable. It marks missing facts instead of inventing them, preserves the source identifier and explains the arithmetic.

This makes a later change easier to handle: if the number of reserved places changes, recompute the result and update the sentence.

**If N2 now reserves five helper places, calculate the remaining places. State which existing source facts stay the same and why the old answer of 21 must be revised**

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STUDENT LESSON · 8 OF 18

## Practise together

Practise Generate, Retrieve, or Calculate? with support, then explain the deciding step in your own words.

GUIDED PRACTICE

### Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Different systems create, look up, or compute answers in different ways.

Use a fresh example to explain **generate**.

**Trace** a new **input** → **process** → **output** example.

Find and repair the error in: **Calculate**d **output**s never need checking.

Invent one edge case and name the safe response.

#### Check with a partner

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

### A closer explanation

Two applications display “21 places remain.” One **trace** shows **retrieval** of N2 followed by a calculator call for 6 × 4 − 3 and a sentence built from the result. The other gives no record or tool **trace**.

Their identical visible answers do not establish identical methods. The second answer may be right, but its appearance does not prove **retrieval** or **calculation** took place.

> **A draft says “Join us Thursday in Room 4 for free snacks; 24 places remain.” Use N2 to identify which claims need correction or removal. Write a corrected factual sentence and state the retrieval and calculation steps that support it**

Check the record yourself and independently recompute 6 × 4 – 3. A displayed **trace** is useful **evidence**, but it should refer to an inspectable record and the actual tool result.

A sentence saying “I checked” is not the same as a check you can inspect. If **evidence** is unavailable, say what remains unverified.

### Practise with less help: Separate a correct sum from a correct answer

Use workbook W2 for the connected example “**Generation retrieval** and **calculation**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your **trace** to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for **Generation retrieval** and **calculation**. Proposed explanation: “Retrieving a record proves it is current”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

> Label each arrow with its operation. Can the system add snacks because invitations often mention them?

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

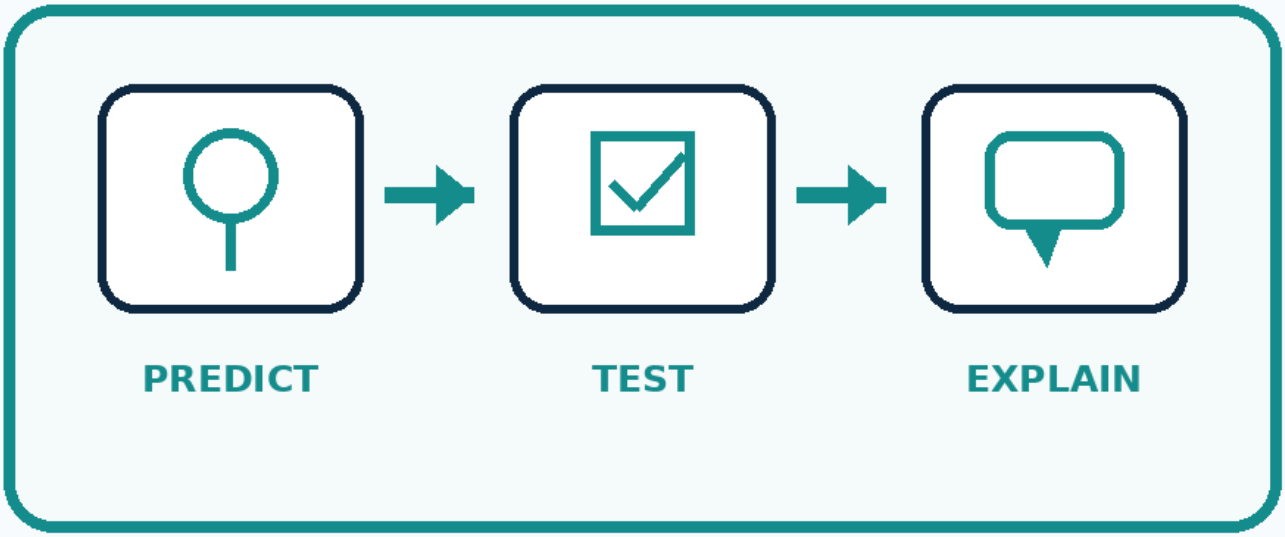
## Run the investigation

> Investigate Generate, Retrieve, or Calculate?. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EXPLORER LAB

### Mechanism Sorting Station

Predict first, test honestly, and explain what the **evidence** means.




### 1. Predict

State your **prediction** before the result is known.

### 2. Investigate

Complete the task: Classify nine **output**s by **mechanism** and verification need.

### 3. Compare cases

Keep a normal case and an edge case clearly separated.

### 4. Record honestly

Record what happened—even when it does not match your **prediction**.

### 5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

| Case | Prediction | Observed result | What it means |
|---|---|---|---|
| Normal | | | |
| Edge | | | |

### Use a fresh case after support

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

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STUDENT LESSON · 10 OF 18

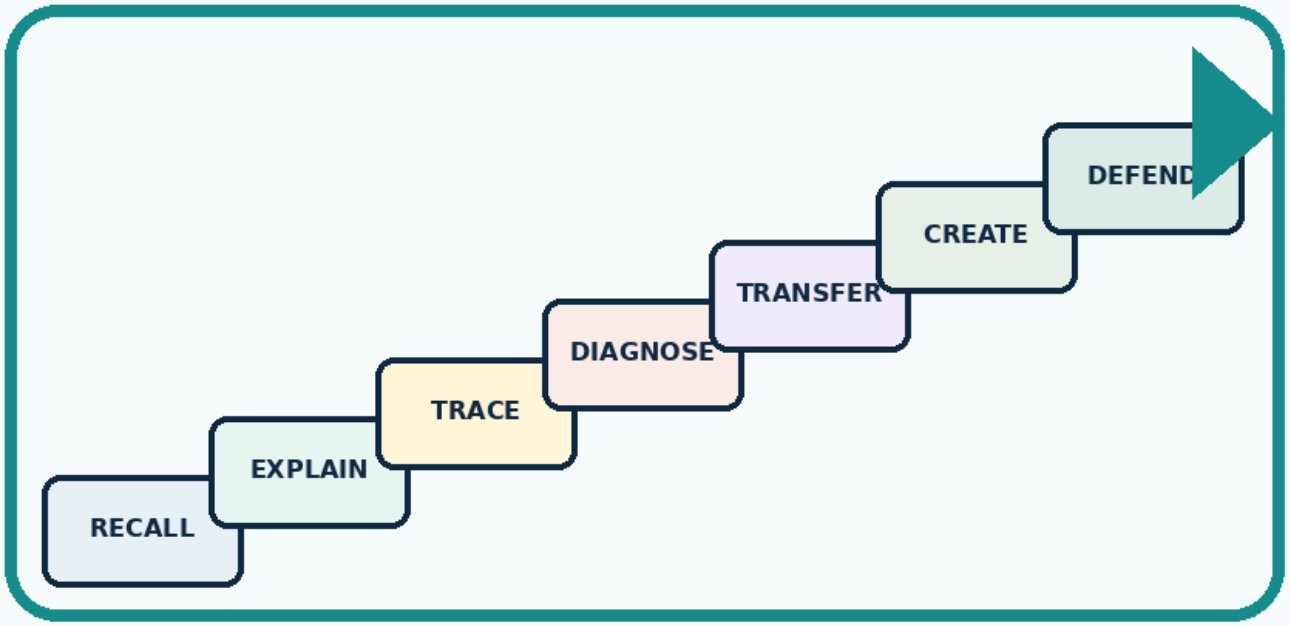
## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Generate, Retrieve, or Calculate?. Explain why each step is justified.

PRACTICE LADDER

**Climb from Recall to Defense**

Each step asks for a stronger kind of understanding.




RECALL Define **generate**.

EXPLAIN Different systems create, look up, or compute answers in different ways.

**TRACE** Show the **input**, **process**, and **output**.

DIAGNOSE Repair: Any answer on a screen came from a database.

TRANSFER Apply the idea to a new home or classroom case.

CREATE Design one revealing test.

DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

### When to retry

RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

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STUDENT LESSON · 11 OF 18

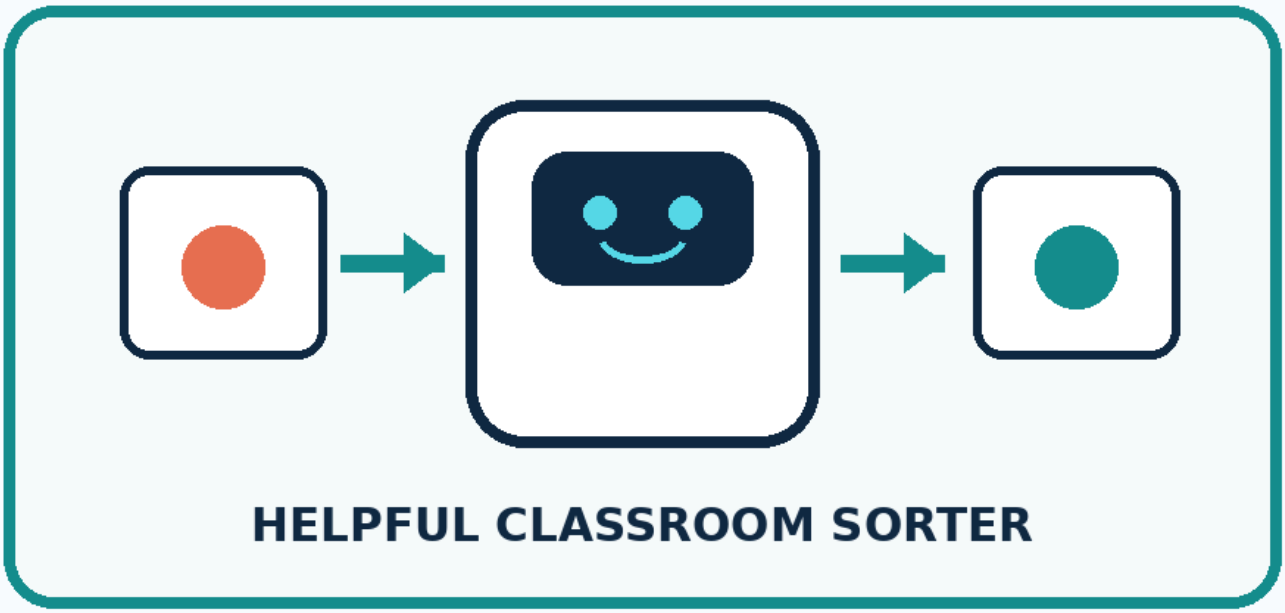
## Connect your project

> Apply this week's idea to Helpful Classroom Sorter. Keep an annotated decision and the evidence supporting it.

PROJECT

### Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.




THIS WEEK'S CHECKPOINT Label every sorter component as **generate**, **retrieve**, **calculate**, fixed rule, learned **prediction**, or **human decision**.

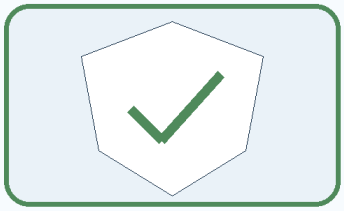
What real classroom need does this serve?

What information is necessary—and what should stay private?

What happens inside the **process**?

What normal and edge cases will you test?

When should it say not sure or ask a person?



### Responsible design

RESPONSIBLE DESIGN Fluent wording does not reveal whether a fact was **retrieve**d, **generate**d, or **calculate**d.

My project decision:

**Evidence for your project**

Use the sorter’s approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations.

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STUDENT LESSON · 12 OF 18

## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

MASTERY

### Fresh-Case Independent Proof

Complete this after teaching and guided practice have ended.




#### What success looks like

REQUIRED PROOF Classify nine **output**s as **generate**d, **retrieve**d, **calculate**d, or mixed and assign an appropriate verification step.

- ☐ Name or draw the fresh case.
- ☐ **Trace input**, **process**, and **output**.
- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Independent **evidence**:

### Check your explanation

SELF-CHECK Check **generate**d claims against independent **evidence**, check **retrieve**d facts against the cited record and freshness, and recompute **calculation**s or inspect the operation. Mixed **output**s need more than one check.

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

### Connect to the next lesson

NEXT BRIDGE Carry today's **evidence** habit into: Prompts Are Design Instructions.

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STUDENT LESSON · 13 OF 18

# Study a complete case

> Revisit the complete Generate, Retrieve, or Calculate? case for supported practice. Repeated examples do not become fresh assessments.

### Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

A learner needs a new poem, 18+7, and a passage from the class handbook.

### The question

Choose the operation for each.

### Worked explanation

**Generate** the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

### Explain it in your own words

> Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

### Trace the supplied case visually

### Choose the operation that fits

| Evidence or stage | Value or result |
|---|---|
| New rain poem | Generate |
| 18 + 7 | Calculate → 25 |
| Handbook passage | Retrieve the actual text |

A learner needs a new poem, 18+7, and a passage from the class handbook.

Choose the operation for each.

**Generate** the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Trace a new library workshop

Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3.

A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” **Retrieve** the current day and room, **calculate** the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Choose a method for each task: invent a dragon story; find the library's posted hours; compute 7 plus 5. Explain why the methods differ.

My answer, with a **trace**, **calculation** or supporting **evidence**

> **Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

### Connect to your project

Label every sorter component as **generate**, **retrieve**, **calculate**, fixed rule, learned **prediction**, or **human decision**.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

### Transfer practice from the original programme

> Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose **generation**, **retrieval** or **calculation** for each. Which **output**s need checking?

> **Explain the mechanism, evidence and one limit.**

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

> Use this map to navigate Generate, Retrieve, or Calculate?, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional craft invitation combines three different operations.

| By the end, I can | How I will show it |
|---|---|
| Distinguish retrieving a record, calculating from inputs and generating wording | Explain the deciding step, show a trace or test, and state one limit. |
| Trace a task that combines these operations and check each step | Explain the deciding step, show a trace or test, and state one limit. |
| Identify unsupported details and explain why output appearance does not reveal its method | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Generate, Retrieve, or Calculate? without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Distinguish retrieving a record, calculating from **input**s and generating wording

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional craft invitation combines three different operations.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? A sentence promising snacks has no supporting record.

1. **Retrieve** facts

2. **Calculate**

3. **Generate** wording

4. Check claims

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Check claims A sentence promising snacks has no supporting record.

**Retrieval** copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Reserve four places instead of two, keeping the room and day unchanged. Which conclusion is supported by the supplied facts?

1. Eight learner places remain.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Eight learner places remain. 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

#### Repair a missing building block

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

Review Similarity (ai-1/19). Goal: **Calculate** similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C′s new score and explain why there is no tie..

Return to this week and explain what you changed.

#### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

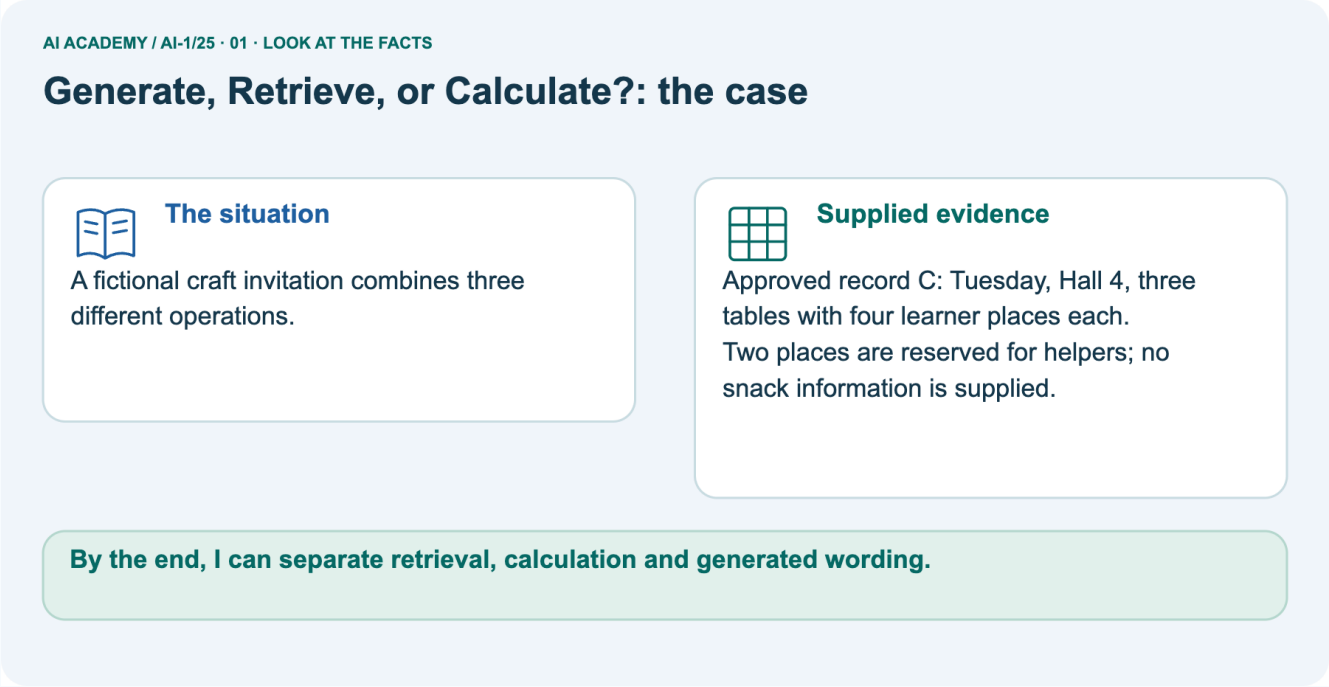
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

# Visual concept journey

> Explain each Generate, Retrieve, or Calculate? graphic using its labels, deciding evidence and relevant limitation.

### 01. Generate, Retrieve, or Calculate?: the case



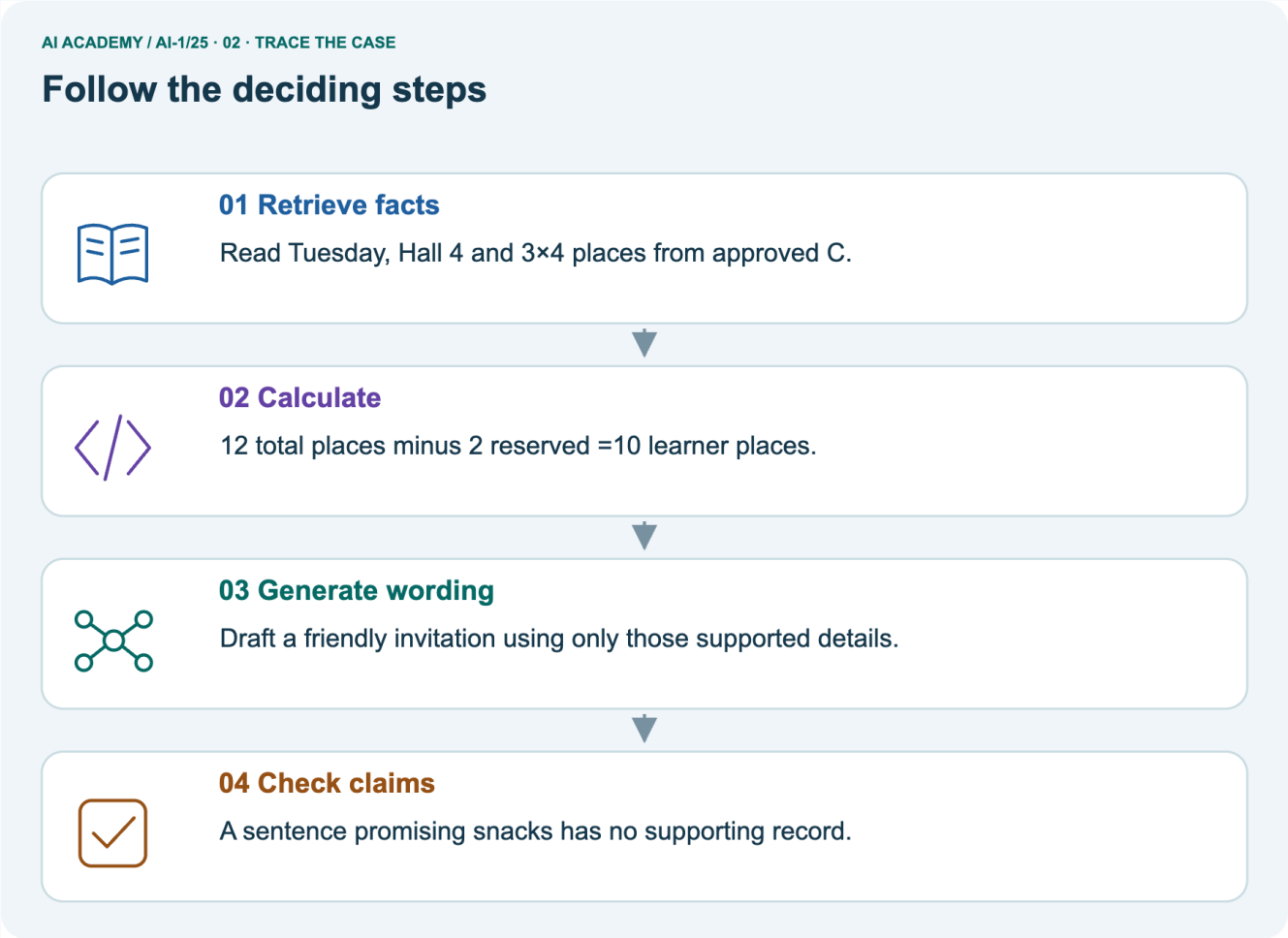

Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft invitation combines three different operations. Supplied facts: Approved record C: Tuesday, Hall 4, three tables with four learner places each.

Two places are reserved for helpers; no snack information is supplied.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

**Retrieve** facts: Read Tuesday, Hall 4 and 3×4 places from approved C. → **Calculate**: 12 total places minus 2 reserved =10 learner places. → **Generate** wording: Draft a friendly invitation using only those supported details. → Check claims: A sentence promising snacks has no supporting record.

> Point to each numbered stage and explain which recorded **input** it uses.

### 03. One club task with three kinds of work

AI ACADEMY / AI-1/25 · 03 · INSPECT THE LESSON MODEL

## One club task with three kinds of work

| Operation | Example output | Required check |
|---|---|---|
| Retrieve N2 | Room 4 on Wednesday | Right event and version |
| Calculate | 6 × 4 − 3 = 21 | Inputs and operations |
| Generate wording | Try a making activity | Facts and constraints |

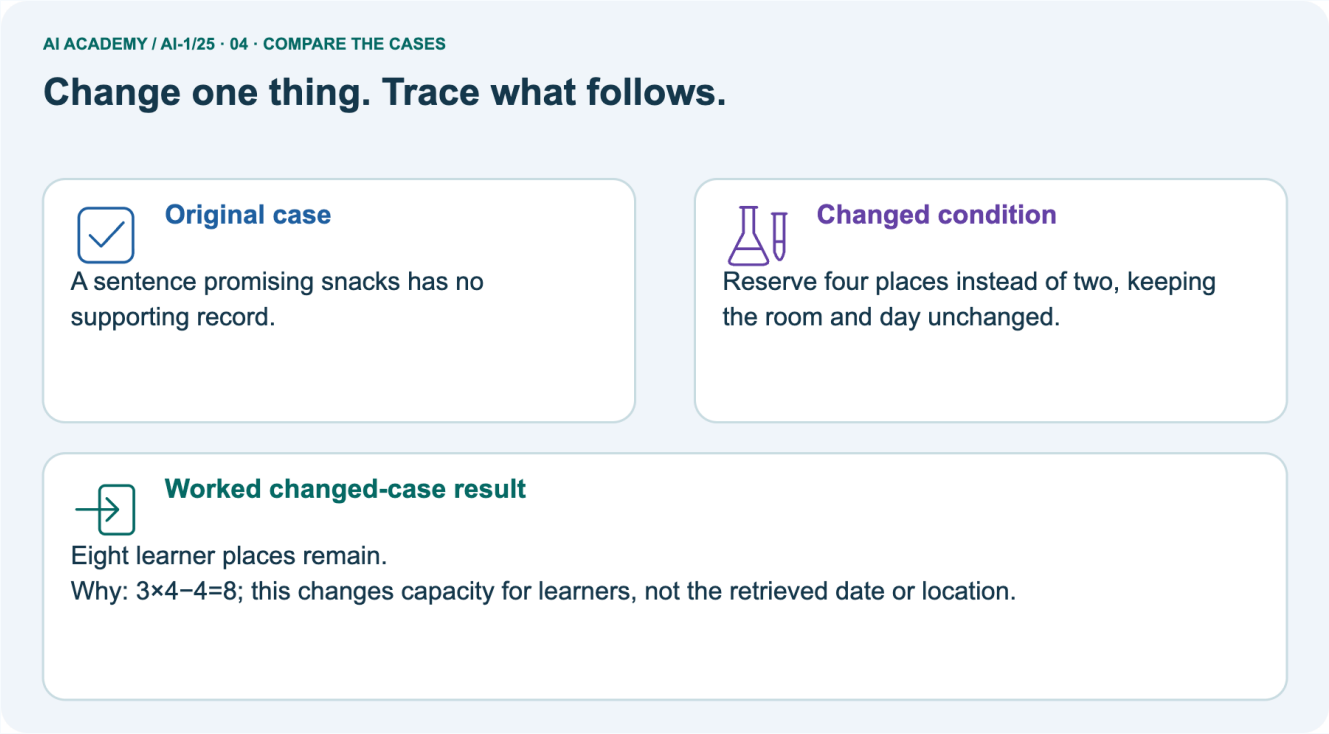
Retrieval supplies recorded information; calculation derives a result; generation drafts wording. Each operation needs a check suited to what it does.

Follow the labelled picture with the matching explanation. Open the vector diagram

One club task with three kinds of work. **Retrieve** N2 | Room 4 on Wednesday | Right event and version. **Calculate** | 6 × 4 − 3 = 21 | **Input**s and operations. **Generate** wording | Try a making activity | Facts and constraints

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



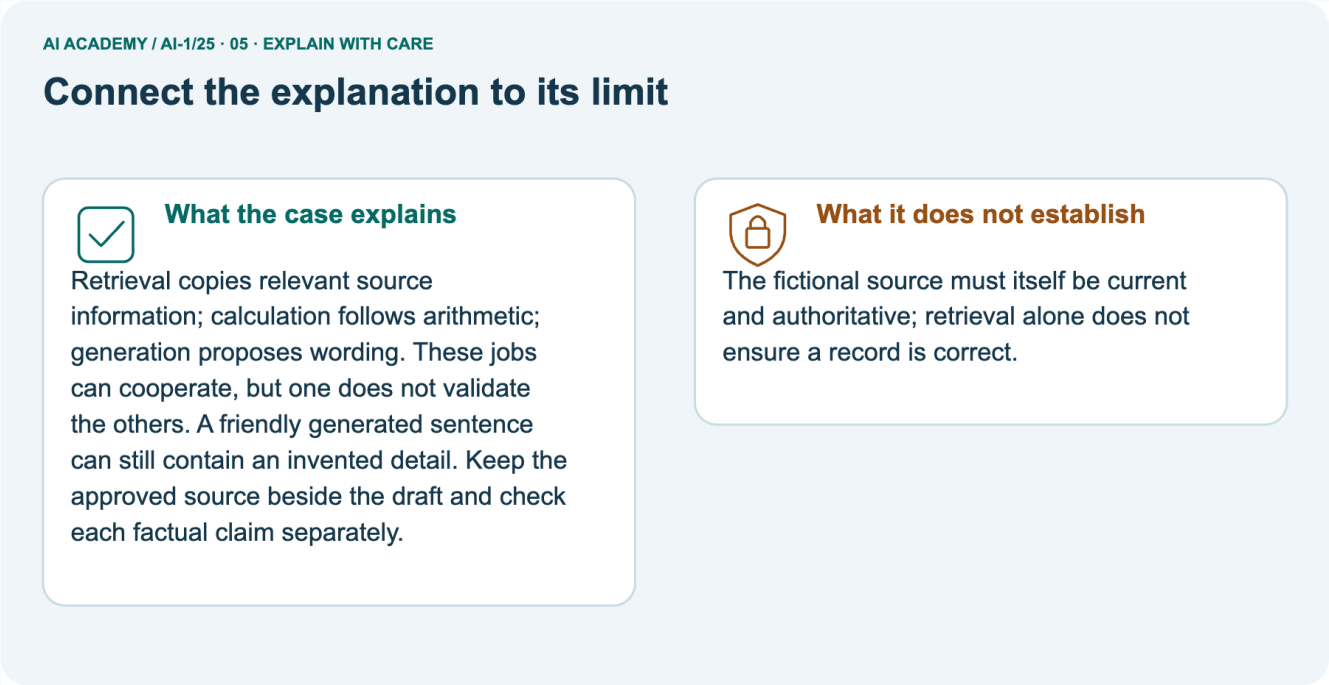

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A sentence promising snacks has no supporting record. Changed condition: Reserve four places instead of two, keeping the room and day unchanged. Expected result: Eight learner places remain.

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

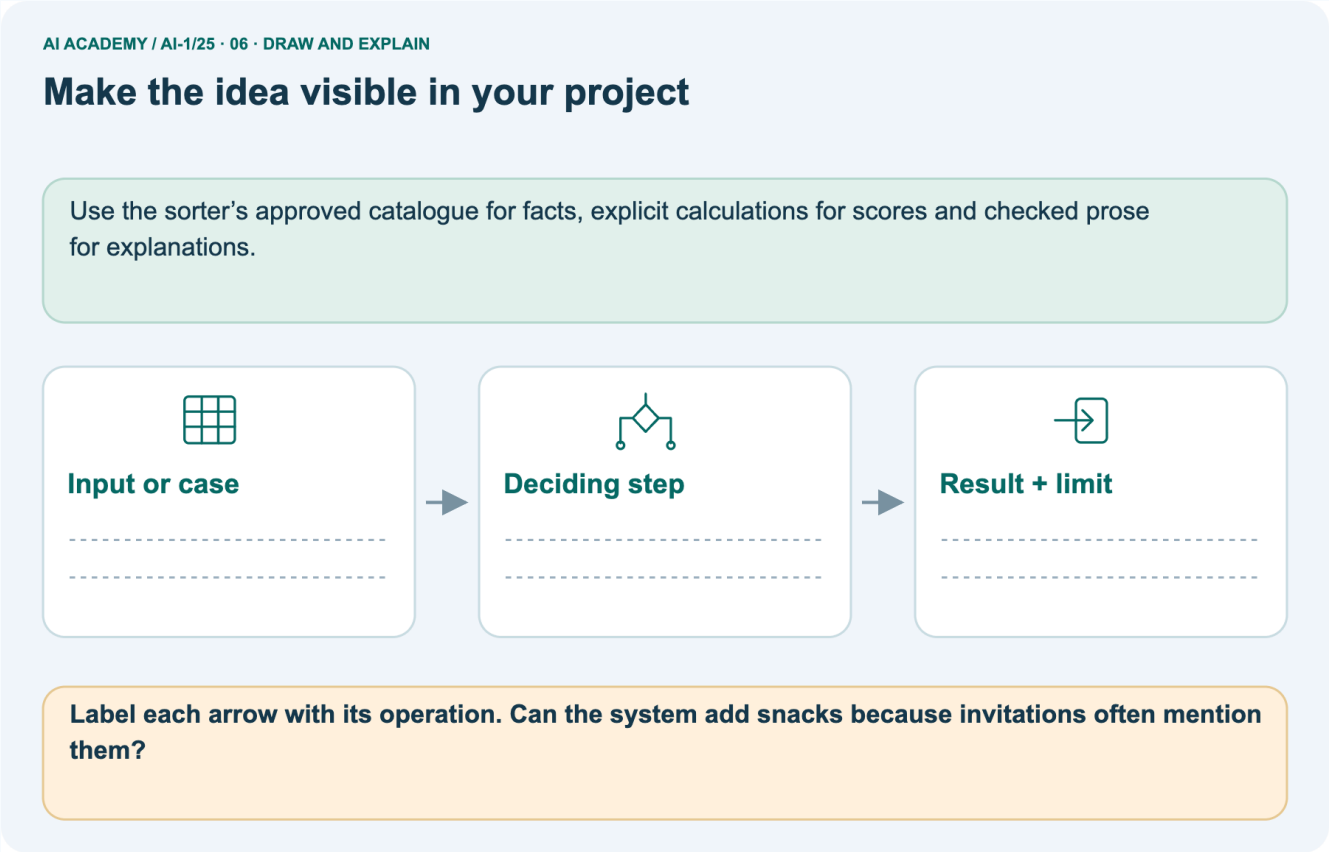
**Retrieval** copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

Limit: The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Use the sorter's approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations. Draw your **input**, deciding step, result and limit. Supported question: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?

> Draw your project **trace** and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.